

Calculation of quantum eigens with geometrical algebra rotors

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Introduction to geometric algebra

"...for I cannot believe that anything so ugly as the multiplication of matrices is an essential part of the scheme of nature".

sir Arthur Eddington

Vector algebra

- Coordinate free
- Has geometric meaning
- Works only in Euclidian 3D space+inconsistent

Tensor method

- Coordinate method
- No geometric meaning
- Works in all dimensions and signatures

Geometric algebra

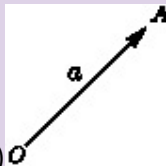
- Coordinate free
- Has geometric meaning
- Works in all dimensions and signatures

Introduction to geometric algebra

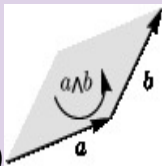
Seminar goal: to solve quantum mechanical eigenvalue/eigenvector problem **without explicitly using matrices** i. e. no diagonalization procedure.

Geometric algebra objects

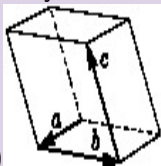
GA deals with oriented object. These objects has orientation in space (called attitude) an internal orientation. Examples: scalar, vectors, bivectors, trivectors, etc..., totally 2^n elements



a)



b)



c)

- a) Translation vector a is an oriented line segment, which size is $|a| = OA$. What about radius vectors?
- b) Bivector $a \wedge b$ is oriented part of plane, size is area, shape not important.
- c) Trivector (pseudoscalar) $a \wedge b \wedge c$ is an oriented volume, orientation is defined by left/right coordinate system made of a , b or c , size is volume, shape not important

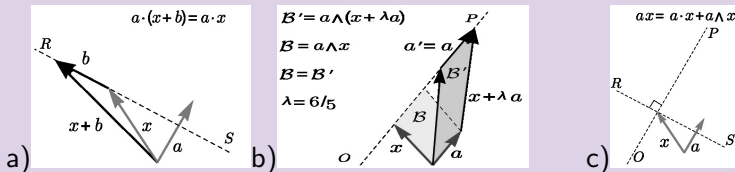
Introduction to geometric algebra

How to make oriented objects from simpler one? Need a product.

Geometric algebra products

GA has 3 types of products:

- outer product ($\wedge$), fully antisymmetric, associative [vector product is not!, i.e. $(a \times b) \times c \neq a \times (b \times c)$], however $|a \wedge b| \leq |a| |b|$.
- inner product ($\cdot$), analog of vector scalar product (few types)
- geometric product (space), associative, includes information from both products above, preserves norm $|ab| = |a| |b|$ (very usefull).



- a) Inner product $a \cdot x$ doesn't change if parallel to b vector is added (freedom is represented by line $RS \perp a$).
- b) Outer product $a \wedge x$ doesn't change if perpendicular to x vector a is added. (freedom is represented by line $OP \parallel a$).
- c) Only knowing both product allows finding unique x , where **both** lines $OP \parallel a$ and $RS \perp a$ intersect. Allows **inverse** of ∇a , whereas separately $\text{div } a = \nabla \cdot a$ and $\text{curl } a = \nabla \times a$ do not!

Introduction to geometric algebra

Associative algebra = Sum + associative Product

Clifford algebra $\iff$ geometric product

$$\begin{aligned} \mathbf{a} \mathbf{b} &= \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}, \\ \mathbf{b} \mathbf{a} &= \mathbf{a} \cdot \mathbf{b} - \mathbf{a} \wedge \mathbf{b}. \end{aligned} \quad \text{GEOMETRIC PRODUCT (of vectors)}$$

Solving for product terms results in

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \frac{1}{2}(\mathbf{a} \mathbf{b} + \mathbf{b} \mathbf{a}), \\ \mathbf{a} \wedge \mathbf{b} &= \frac{1}{2}(\mathbf{a} \mathbf{b} - \mathbf{b} \mathbf{a}). \end{aligned}$$

If vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel, then $\mathbf{a} \wedge \mathbf{b} = 0$, therefore $\mathbf{a} \mathbf{b} = \mathbf{b} \mathbf{a}$, if $\mathbf{a} \parallel \mathbf{b}$. Taking $\mathbf{a} = \mathbf{b}$, have $\mathbf{a} \cdot \mathbf{a} = \mathbf{a}^2 = |\mathbf{a}|^2$. Whereas $\mathbf{a} \wedge \mathbf{a} = 0$.

If vectors perpendicular, then, $\mathbf{a} \cdot \mathbf{b} = 0$, therefore $\mathbf{a} \mathbf{b} = -\mathbf{b} \mathbf{a}$, if $\mathbf{a} \perp \mathbf{b}$.

Conclusion

- Vectors commute $\iff$ they are parallel.
- Vectors anticommute $\iff$ they are perpendicular.

Introduction to geometric algebra

Base vectors of $Cl_{p,q}$

Vectors are of **special** importance.

$$\mathbf{e}_i \mathbf{e}_j + \mathbf{e}_j \mathbf{e}_i = \pm 2\delta_{ij}, \quad i, j = 1, 2, \dots, p + q.$$

Sign in $+$ for $i \leq p$ and $-$ for $p < i \leq p + q$.

For example, for $Cl_{3,0}$ $\mathbf{e}_1^2 = \mathbf{e}_2^2 = \mathbf{e}_3^2 = 1$, signature $(+++)$.

For Minkowski space $Cl_{1,3}$, $\mathbf{e}_1^2 = 1, \mathbf{e}_2^2 = \mathbf{e}_3^2 = \mathbf{e}_4^2 = -1$.

$Cl_{1,3} \not\cong Cl_{3,1}$ (!) Algebras aren't isomorphic.

Base vectors are orthonormal, $\mathbf{e}_i \cdot \mathbf{e}_j = 0$, if $i \neq j$, therefore their outer product coincides with geometric product

$\mathbf{e}_i \mathbf{e}_j = \mathbf{e}_i \wedge \mathbf{e}_j = -\mathbf{e}_j \wedge \mathbf{e}_i = -\mathbf{e}_j \mathbf{e}_i$. In $Cl_{3,0}$ can make only 3 independent pairs, i. e. $\mathbf{e}_1 \wedge \mathbf{e}_2$, $\mathbf{e}_2 \wedge \mathbf{e}_3$ and $\mathbf{e}_3 \wedge \mathbf{e}_1$. They square to

$$(\mathbf{e}_1 \wedge \mathbf{e}_2)^2 = (\mathbf{e}_2 \wedge \mathbf{e}_3)^2 = (\mathbf{e}_3 \wedge \mathbf{e}_1)^2 = -1.$$

Introduction to geometric algebra

Matrix representations for $Cl_{2,0}$

$$\mathbf{e}_i \mathbf{e}_j + \mathbf{e}_j \mathbf{e}_i = +2\delta_{ij}, \quad i, j = 1, 2, \dots, p + q.$$

$$\mathbf{e}_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Squares gives identity matrix = scalar element. Multiplication of above matrices gives $\mathbf{e}_{12}$ element

$$1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{e}_{12} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Matrix representations has **no grade** information. Note about $Cl_{3,0}$ invariants: energy (scalar), momentum (vector), angular momentum (bivector), ...

Introduction to geometric algebra

Matrix representations for $Cl_{1,1}$

$$\mathbf{e}_i \mathbf{e}_j + \mathbf{e}_j \mathbf{e}_i = \pm 2\delta_{ij}, \quad i, j = 1, 2, \dots, p + q.$$

$$1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{e}_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \mathbf{e}_{12} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Most general element $Cl_{1,1}$ can be written as

$$a_0 + a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_{12} = \begin{bmatrix} a_0 + a_1 & a_2 + a_3 \\ -a_2 + a_3 & a_0 - a_1 \end{bmatrix}.$$

Introduction to geometric algebra

Matrix representations for $Cl_{0,2}$ (quaternions)

$$\mathbf{e}_i \mathbf{e}_j + \mathbf{e}_j \mathbf{e}_i = -2\delta_{ij}, \quad i, j = 1, 2, \dots, p + q.$$

$$q = \begin{bmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{bmatrix}.$$

Taking one of q_i equal 1, others 0, we find **complex** matrix representations

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}.$$

When multiplied by $-i$, they coincide with Pauli matrices $\hat{\sigma}_z$, $\hat{\sigma}_y$ and $\hat{\sigma}_x$.

For real representations of quaternions we need 4×4 matrices.

Introduction to geometric algebra

Trigonometric expression for inner product

Useful when calculating without explicit base.

In $Cl_{2,0}$ for inner product

$$a \cdot b = |a||b| \cos \varphi,$$

φ is an angle from a to b in anticlockwise direction,

$|a| = \sqrt{a_1^2 + a_2^2}$ is vector norm (length). Can be calculated as

$|a| = \sqrt{a\tilde{a}}$, Tilde means reversion,

$$\widetilde{abc} = cba.$$

For single vector $\tilde{a} = a$.

Introduction to geometric algebra

Trigonometric expression for outer product

Outer product in coordinate form:

$$a \wedge b = (a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2) \wedge (b_1 \mathbf{e}_1 + b_2 \mathbf{e}_2),$$

$$a \wedge b = a_1 b_1 (\mathbf{e}_1 \wedge \mathbf{e}_1) + a_1 b_2 (\mathbf{e}_1 \wedge \mathbf{e}_2) + a_2 b_1 (\mathbf{e}_2 \wedge \mathbf{e}_1) + a_2 b_2 (\mathbf{e}_2 \wedge \mathbf{e}_2).$$

Because $\mathbf{e}_1 \wedge \mathbf{e}_1 = \mathbf{e}_2 \wedge \mathbf{e}_2 = 0$ and $\mathbf{e}_1 \wedge \mathbf{e}_2 = -\mathbf{e}_2 \wedge \mathbf{e}_1$, have

$$a \wedge b = (a_1 b_2 - a_2 b_1) \mathbf{e}_1 \wedge \mathbf{e}_2 = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{e}_1 \wedge \mathbf{e}_2.$$

Scalar $a_1 b_2 - a_2 b_1 = S$ is area $S = |a||b| \sin \varphi$ of parallelogram, made of vectors a and b . Then it is clear

$$a \wedge b = S \mathbf{e}_{12} = |a||b| \sin \varphi \mathbf{e}_{12}.$$

Introduction to geometric algebra

Trigonometric expression for outer product (other way)

Using $ab = a \cdot b + a \wedge b$ ir $ba = a \cdot b - a \wedge b$, can write

$$abba = (a \cdot b)^2 - (a \wedge b)^2.$$

$bb = \tilde{b}b = |b|^2$ is scalar, which commutes with a . This gives expression for $(a \wedge b)^2$

$$(a \wedge b)^2 = -|a|^2|b|^2(1 - \cos^2 \varphi) = -|a|^2|b|^2 \sin^2 \varphi.$$

Because $e_{12}^2 = -1$ we have the same result.

Introduction to geometric algebra

Exponent of bivector

From inner and outer expressions can write

$$ab = a \cdot b + a \wedge b = |a||b|(\cos \varphi + \mathbf{e}_{12} \sin \varphi).$$

Define exponent

$$\cos \varphi + \mathbf{e}_{12} \sin \varphi = e^{\mathbf{e}_{12}\varphi}.$$

Check expanding as series. Using $\mathbf{e}_{12}^2 = -1$, write as a sum of even and odd series of $\mathbf{e}_{12}$ powers

$$e^{\mathbf{e}_{12}\varphi} = \sum_{n=0}^{\infty} \frac{(\mathbf{e}_{12}\varphi)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n \varphi^{2n}}{(2n)!} + \mathbf{e}_{12} \sum_{n=0}^{\infty} \frac{(-1)^{n+1} \varphi^{2n+1}}{(2n+1)!}.$$

Even series gives $\cos \varphi$, odd is $\sin \varphi$ series. So, $ab = |a||b|e^{\mathbf{e}_{12}\varphi}$.
Reminder for complex numbers $e^{i\varphi} = \cos \varphi + i \sin \varphi$.

Introduction to geometric algebra

Example: derivation formula for $\sin(\alpha + \beta)$

From inner and outer expressions can write

$$e^{\mathbf{e}_{12}\alpha} e^{\mathbf{e}_{12}\beta} = e^{\mathbf{e}_{12}(\alpha+\beta)}$$

$$\begin{aligned} e^{\mathbf{e}_{12}\alpha} e^{\mathbf{e}_{12}\beta} &= (\cos \alpha + \mathbf{e}_{12} \sin \alpha)(\cos \beta + \mathbf{e}_{12} \sin \beta) \\ &= \cos \alpha \cos \beta - \sin \alpha \sin \beta + \mathbf{e}_{12}(\sin \alpha \cos \beta + \sin \beta \cos \alpha) \end{aligned}$$

$$e^{\mathbf{e}_{12}(\alpha+\beta)} = \cos(\alpha + \beta) + \mathbf{e}_{12} \sin(\alpha + \beta)$$

Comparing terms near scalar and bivector $\mathbf{e}_{12}$ we see that

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \text{ and}$$

$$\sin(\alpha + \beta) = (\sin \alpha \cos \beta + \sin \beta \cos \alpha).$$

Introduction to geometric algebra

Rotor

When a and b are of unit length ($|a| = |b| = 1$), their geometric product is exponent of bivector

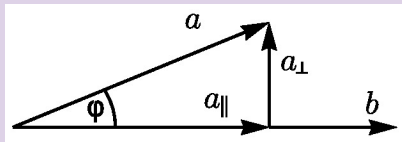
$$\hat{a}\hat{b} \equiv R(\varphi) = e^{e_{12}\varphi}, \quad e_{12}^2 = -1. \quad Cl_{2,0} \text{ ROTOR}$$

They are very similar to complex exponents $e^{i\varphi}$, therefore plays an important role in rotations. Rotor is a **scalar+bivector**

Rotation **axes** is misleading concept. Only planes of rotation, i. e. planes where rotation takes place, are important. Example 2D or 4D (Lorentz transform is a rotation in $Cl_{1,3}$).

Introduction to geometric algebra

Orthogonal projection



Vector a orthogonal projection $a_{\parallel}$ and rejection $a_{\perp}$ with respect to vector b

Vector a projection $a_{\parallel}$ onto vector b . Unit vector is $\hat{b} = b/|b|$. Norm of a projection $|a_{\parallel}| = |a| \cos \varphi$. Therefore

$$a_{\parallel} = |a| |\hat{b}| = |a| |b| (b/|b|^2) \cos \varphi.$$

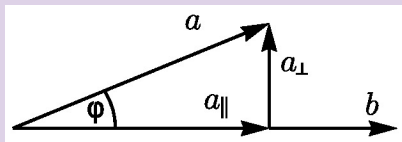
Using $b/|b|^2 = b^{-1}$ and $|a| |b| \cos \varphi = a \cdot b$ yields

$$a_{\parallel} = (a \cdot b) b^{-1}. \quad \text{PROJECTION}$$

Note, that projection is independent on vector b length, therefore, $a_{\parallel} = (a \cdot \hat{b}) \hat{b}^{-1} = (a \cdot \hat{b}) \hat{b}$. Can project to any multivector.

Introduction to geometric algebra

Orthogonal rejection



Vector $a_{\perp}$ is called rejection.

$$a_{\perp} = a - a_{\parallel} = a - (a \cdot b)b^{-1} = (ab - a \cdot b)b^{-1}.$$

Term in brackets is outer product, therefore

$$a_{\perp} = (a \wedge b)b^{-1}. \quad \text{REJECTION}$$

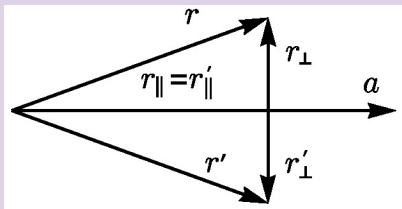
Summing projection and rejection yields initial vector a

$$a_{\parallel} + a_{\perp} = (a \cdot b)b^{-1} + (a \wedge b)b^{-1} = (a \cdot b + a \wedge b)b^{-1} = abb^{-1} = a.$$

Projection and rejection are **vectors**, while in traditional vector algebra they are understood as scalars. Projection and rejection vectors are orthogonal $a_{\parallel} \cdot a_{\perp} = \frac{1}{2}(a_{\parallel}a_{\perp} + a_{\perp}a_{\parallel}) = 0$.

Introduction to geometric algebra

Reflection



Reflection of r with respect to vector a

Parallel component remains unchanged, while that orthogonal to a change sign

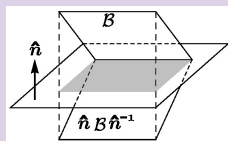
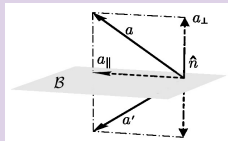
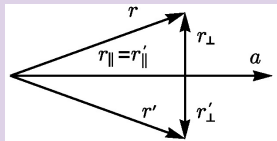
$$r' = r'_{\parallel} + r'_{\perp} = r_{\parallel} - r_{\perp}.$$

Inserting expression for projection $r_{\parallel}$ and rejection $r_{\perp}$ have

$$\begin{aligned} r' &= (r \cdot a)a^{-1} - (r \wedge a)a^{-1} \\ &= (r \cdot a - r \wedge a)a^{-1} = (a \cdot r + a \wedge r)a^{-1} \\ &= ara^{-1}. \end{aligned}$$

Introduction to geometric algebra

Reflection (continuation)



Actually important is only vector a , direction, because $a = |a|\hat{a}$ and $a^{-1} = \hat{a}/|a|$, therefore scalar $|a|$ will disappear. So

$$r' = \hat{a} r \hat{a}. \quad \text{REFLECTION}$$

This is one of most beautiful formula in GA. For bivector reflection in 3D (note the minus sign)

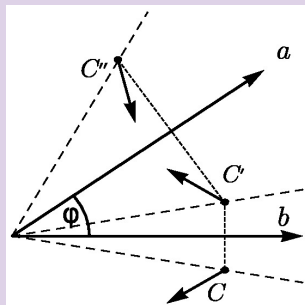
$$\mathcal{B}' = (-\hat{n} a \hat{n}) \wedge (-\hat{n} b \hat{n}).$$

And simplify

$$\mathcal{B}' = \frac{1}{2}(\hat{n} a \hat{n} \hat{n} b \hat{n} - \hat{n} b \hat{n} \hat{n} a \hat{n}) = \frac{1}{2} \hat{n} (ab - ba) \hat{n} = \hat{n} \mathcal{B} \hat{n}.$$

Introduction to geometric algebra

Combination of $2n$ reflections = rotation

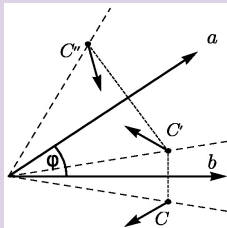


Two reflections, one with respect to b , other with respect to a .
 $C \rightarrow C' \rightarrow C''$. First reflection: $r \rightarrow r' = \hat{b}r\hat{b}^{-1}$. Second reflection $r' \rightarrow r'' = \hat{a}r'\hat{a}^{-1}$. Composition results in

$$\begin{aligned} r'' &= \hat{a}(\hat{b}r\hat{b}^{-1})\hat{a}^{-1} = (\hat{a}\hat{b})r(\hat{b}^{-1}\hat{a}^{-1}) \\ &= (\hat{a}\hat{b})r(\hat{a}\hat{b})^{-1}. \end{aligned}$$

Introduction to geometric algebra

Rotation



Rewrite using formula for exponent

$$r'' = e^{\mathbf{e}_{12}\varphi} r (e^{\mathbf{e}_{12}\varphi})^{-1}.$$

$\exp(-\mathbf{e}_{12}\varphi) \exp(\mathbf{e}_{12}\varphi) = (\cos \varphi - \mathbf{e}_{12} \sin \varphi)(\cos \varphi + \mathbf{e}_{12} \sin \varphi) = 1$,
then $(e^{\mathbf{e}_{12}\varphi})^{-1} = e^{-\mathbf{e}_{12}\varphi}$ and can rewrite as

$$r'' = e^{\mathbf{e}_{12}\varphi} r e^{-\mathbf{e}_{12}\varphi}.$$

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Standard quantum mechanics approach

Two level model

The Hamiltonian

$$\hat{H} = \begin{bmatrix} \varepsilon_0 - \varepsilon_3 & \varepsilon_1 - i\varepsilon_2 \\ \varepsilon_1 + i\varepsilon_2 & \varepsilon_0 + \varepsilon_3 \end{bmatrix},$$

where ε_i are reals. Pauli matrices

$$\hat{\sigma}_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \hat{\sigma}_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \hat{\sigma}_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

are $Cl_{3,0}$ vector representations, because

$$\hat{\sigma}_i \hat{\sigma}_j + \hat{\sigma}_j \hat{\sigma}_i = \begin{cases} 0, & \text{jei } i \neq j \\ 2\hat{1}, & \text{jei } i = j \end{cases}, \quad \text{where } \hat{1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

and $\hat{\sigma}_1^2 = \hat{\sigma}_2^2 = \hat{\sigma}_3^2 = \hat{1}$. Solution is trivial.

Standard quantum mechanics approach

Electron in a monolayer graphene

In the Hilbert space formalism, the **massless Dirac Hamiltonian** in the vicinity of critical K point of graphene reduces to simple form

$$\hat{H}_{MG} = \hbar v_F \begin{bmatrix} 0 & k_x - i k_y \\ k_x + i k_y & 0 \end{bmatrix}, \quad (1)$$

where $v_F \approx 10^6$ m/s is the Fermi velocity. The wave vectors $\mathbf{k} = (k_x, k_y)$ lies in the graphene plane and is measured with respect to valley minimum. In the following the natural units $\hbar = 1$ along with $v_F = 1$ are used.

Standard quantum mechanics approach

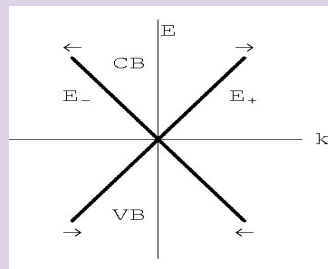
Eigenenergies of monolayer graphene

The eigenenergies of the Hamiltonian (1) are

$$E_{\pm} = \pm \sqrt{k_x^2 + k_y^2} = \pm |\mathbf{k}|,$$

where plus/minus sign corresponds to conduction/valence bands. Since $E \propto |\mathbf{k}|$, the spectrum is ultrarelativistic.

Spectrum of monolayer graphene



CB/VB means conduction/valence band. Left/right arrows show pseudospin direction.

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$Cl_{3,0}$ algebra approach

Projection into base matrices

Calculate products of two matrices $\hat{\sigma}_1^2 = \hat{\sigma}_2^2 = \hat{\sigma}_3^2 = \hat{1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

and

$$\hat{\sigma}_1\hat{\sigma}_2 = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \quad \hat{\sigma}_2\hat{\sigma}_3 = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}, \quad \hat{\sigma}_3\hat{\sigma}_1 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

When trace is nonzero? From the Hamiltonian

$$\hat{H} = \begin{bmatrix} \varepsilon_0 + \varepsilon_3 & \varepsilon_1 - i\varepsilon_2 \\ \varepsilon_1 + i\varepsilon_2 & \varepsilon_0 - \varepsilon_3 \end{bmatrix},$$

ε_i can be extracted using

$$\varepsilon_i = \frac{1}{d} \text{Tr}(\hat{H} \hat{\sigma}_i).$$

$Cl_{3,0}$ algebra approach

Algebraic structure

The result is

$$\hat{H} = \varepsilon_0 + \varepsilon_1 \hat{\sigma}_1 + \varepsilon_2 \hat{\sigma}_2 + \varepsilon_3 \hat{\sigma}_3 .$$

Any 2×2 hermitian matrix can be expanded in base of

$$\{1, \sigma, I\sigma, I\} = \{\text{scalar, 3 vectors, 3 bivectors, trivector (pseudoscalar)}\}.$$

When formulating problem purely in geometric algebra remove hats from $\hat{\sigma}_i$, i.e.

$$H = \varepsilon_0 + \varepsilon_1 \sigma_1 + \varepsilon_2 \sigma_2 + \varepsilon_3 \sigma_3 .$$

Problem. $\hat{H}\psi = E\psi$. What to do with ψ ?

$Cl_{3,0}$ algebra approach

What is the spinor in GA?

Simplified idea: replace spinor by matrix

$$\begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} \rightarrow \begin{bmatrix} \psi_1 & 0 \\ \psi_2 & 0 \end{bmatrix}$$

An ideal is a subset I of elements in a ring R that forms an additive group and has the property that, whenever x belongs to R and y belongs to I , then xy and yx belong to I . For example, the set of even integers is an ideal in the ring of integers $\mathbb{Z}$. Given an ideal I , it is possible to define a quotient ring R/I .

In geometric algebra spinor is a sum of even elements:
scalar + bivector + $\dots$

$Cl_{3,0}$ algebra approach

What is the spinor in GA? (continuation)

Replacement rules

$$\begin{bmatrix} a_0 + ia_3 \\ -a_2 + ia_1 \end{bmatrix} = |\psi\rangle \quad \Leftrightarrow \quad \psi = a_0 + a_1 I\sigma_1 + a_2 I\sigma_2 + a_3 I\sigma_3 ,$$

$Cl_{3,0}$ SPINOR

where a_i are any real numbers, which satisfy

$$\langle\psi|\psi\rangle = \tilde{\psi}\psi = \psi\tilde{\psi} = a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1.$$

$\tilde{\psi}$ can be replaced by

$$[a_0 - ia_3, -a_2 - ia_1] = \langle\psi| \quad \Leftrightarrow \quad \tilde{\psi}.$$

However for different spinors $\langle\psi|\phi\rangle \neq \tilde{\psi}\phi$.

$Cl_{3,0}$ algebra approach

Replacement rules

More replacement rules

Hilbert space	$Cl_{3,0}$ algebra	Hilbert space	$Cl_{3,0}$ algebra
$\hat{1} \psi\rangle$	ψ	$\hat{\sigma}_k \psi\rangle$	$\sigma_k \psi \sigma_3$
$i \psi\rangle$	$\psi I \sigma_3$	$i \hat{\sigma}_k \psi\rangle$	$I \sigma_k \psi$
$(a_r + i a_i) \psi\rangle$	$a_r \psi + a_i \psi I \sigma_3$	$\partial \psi\rangle/\partial t$	$\dot{\psi}$
$ \psi^*\rangle$	$\sigma_2 \psi \sigma_2$	$i \partial \psi\rangle/\partial t$	$I \dot{\psi} \sigma_3$
$\langle\psi $	$\tilde{\psi}$	$\hat{k}_x \psi\rangle \equiv -i \frac{\partial \psi\rangle}{\partial x}$	$-\frac{\partial\psi}{\partial x} I \sigma_3$

Relation between traditional Hilbert space formalism and $Cl_{3,0}$
geometric algebra

$Cl_{3,0}$ algebra approach

Two level model: reformulation

Using replacement rules $\hat{H}\psi$ is equivalent to

$$H(\psi) = \varepsilon_0\psi + (\varepsilon_1\sigma_1 + \varepsilon_2\sigma_2 + \varepsilon_3\sigma_3)\psi\sigma_3.$$

Note the appearance of privileged direction σ_3 . Rewrite using help vector $\varepsilon = \varepsilon_1\sigma_1 + \varepsilon_2\sigma_2 + \varepsilon_3\sigma_3$,

$$H(\psi) = \varepsilon_0\psi + \varepsilon\psi\sigma_3.$$

Need to solve algebraic equation with noncommuting quantities

$$\varepsilon_0\psi_{\pm} + \varepsilon\psi_{\pm}\sigma_3 = \varepsilon_{\pm}\psi_{\pm}.$$

Rewrite it as an rotor equation, i. e. multiply from left by σ_3 and then by $\psi_{\pm}$:

$$\varepsilon_0\psi_{\pm}\sigma_3\tilde{\psi}_{\pm} + \varepsilon\psi_{\pm}\tilde{\psi}_{\pm} = \varepsilon_{\pm}\psi_{\pm}\sigma_3\tilde{\psi}_{\pm}.$$

$Cl_{3,0}$ algebra approach

Two level model: eigenvalues

Look for spinor satisfying $\psi_{\pm} \tilde{\psi}_{\pm} = 1$. Then above formula simplifies to an elegant form

$$\psi_{\pm} \sigma_3 \tilde{\psi}_{\pm} = \frac{\epsilon}{\epsilon_{\pm} - \epsilon_0}.$$

Due to normalization requirement left side is an rotor, which rotates vector σ_3 into vector ϵ . Length of σ_3 is unity and cannot change under rotation. Therefore square of right hand side is

$$\left(\frac{\epsilon}{\epsilon_{\pm} - \epsilon_0} \right)^2 = \frac{\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2}{(\epsilon_{\pm} - \epsilon_0)^2} = 1.$$

This is the **quantization condition**! After solving it with respect to $\epsilon_{\pm}$ we obtain eigenvalues

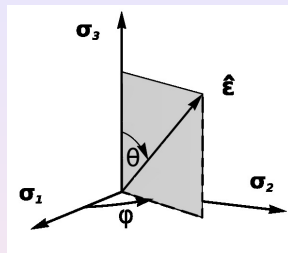
$$\epsilon_{\pm} = \epsilon_0 \pm |\epsilon| = \epsilon_0 \pm \sqrt{\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2}.$$

$Cl_{3,0}$ algebra approach

Two level model: eigenvectors

After eigenvalues are known can solve for eigenvectors ε_+ and ε_- ,

$$\psi_{\pm} \sigma_3 \tilde{\psi}_{\pm} = \pm \frac{\varepsilon}{|\varepsilon|} \equiv \pm \hat{\varepsilon}.$$



Because initial σ_3 and final $\hat{\varepsilon}$ unit vectors are known, the spinors-rotors can be easily constructed

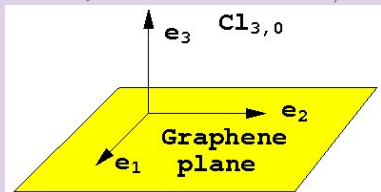
$$\psi_+ = \exp\left(-\frac{\sigma_3 \wedge \hat{\varepsilon}}{|\sigma_3 \wedge \hat{\varepsilon}|} \frac{\theta}{2}\right) = \cos \frac{\theta}{2} - \frac{\sigma_3 \wedge \hat{\varepsilon}}{|\sigma_3 \wedge \hat{\varepsilon}|} \sin \frac{\theta}{2},$$

Here θ denotes angle between vectors σ_3 and $\hat{\varepsilon}$. Rotation angle is from σ_3 to $\hat{\varepsilon}$. $\cos \varphi = \varepsilon_1 / (\varepsilon_1^2 + \varepsilon_2^2)^{1/2}$.

$Cl_{3,0}$ algebra approach

$Cl_{3,0}$ algebra

Minimal algebra to describe the monolayer Hamiltonian is $Cl_{3,0}$.



Base vectors $\mathbf{e}_1^2 = \mathbf{e}_2^2 = \mathbf{e}_3^2 = 1$ play the role of space coordinates. The orthogonal oriented planes are bivectors $\mathbf{Ie}_1 = \mathbf{e}_2\mathbf{e}_3$, $\mathbf{Ie}_2 = \mathbf{e}_3\mathbf{e}_1$, and $\mathbf{Ie}_3 = \mathbf{e}_1\mathbf{e}_2$. With scalar and pseudoscalar $I = \mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$ included, there are 8 basis elements.

Replacement rules

Transform the Hamiltonian (1) to $Cl_{3,0}$ using mapping rules from Hilbert to Clifford space [Doran and Lasenby, 2003]

$$|\psi\rangle \Leftrightarrow \psi = a_0 + a_1\mathbf{Ie}_1 + a_2\mathbf{Ie}_2 + a_3\mathbf{Ie}_3,$$

$$\lambda|\psi\rangle \Leftrightarrow \lambda\psi,$$

$$\hat{\sigma}_i|\psi\rangle \Leftrightarrow \mathbf{e}_i\psi\mathbf{e}_3.$$

λ is the scalar and $\hat{\sigma}_i$ are Pauli matrices.

Then (1), $\hat{H}_{MG}|\psi\rangle$, maps to the following simple GA function of spinor ψ

$$\mathbf{H}_{MG}(\psi) = \mathbf{k}\psi\mathbf{e}_3. \quad (2)$$

The wave vector $\mathbf{k} = k_x\mathbf{e}_1 + k_y\mathbf{e}_2$ lies in the graphene plane. $\mathbf{e}_3$ represents the pseudospin quantization axis, which is perpendicular to graphene plane.

$Cl_{3,0}$ algebra approach

Eigenvalue equation

From (2) eigenvalue equation for monolayer graphene follows

$$\mathbf{k}\psi\mathbf{e}_3 = \psi E,$$

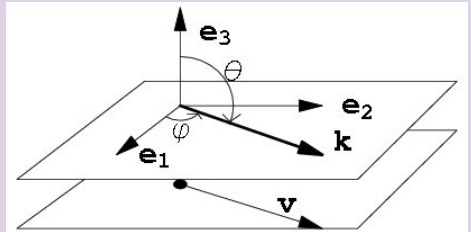
which can be easily rewritten as an equation for rotor

$$\frac{\psi\mathbf{e}_3\tilde{\psi}}{\psi\tilde{\psi}} = \frac{\mathbf{k}}{|\mathbf{k}|^2}E, \quad (3)$$

where $|\mathbf{k}| \equiv k = \sqrt{k_x^2 + k_y^2}$ is the magnitude of the wave vector. Since $\mathbf{e}_3$ is a unit vector, after rotation by ψ we shall obtain the unit vector on the right hand side too.

Eigenvalues

This means that the square of right hand side of (3) must be 1, $(E/k)^2 = 1$, from which follows 2D ultrarelativistic dispersion law $E_{\pm} = \pm k = \pm\sqrt{k_x^2 + k_y^2}$.



Electron with velocity $\mathbf{v}$ and coordinate system for graphene. The planes represent graphene layers. $\mathbf{k}$ is electron wave vector. $\theta = \pi/2$, $\cos \varphi = k_1/|\mathbf{k}|$, $\sin \varphi = k_2/|\mathbf{k}|$.

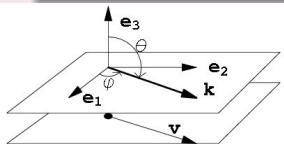
$Cl_{3,0}$ algebra approach

Eigenmultivectors of monolayer graphene

The corresponding eigenmultivectors can be deduced from the same equation (3) after insertion of eigenvalue $E_{\pm}$ back and requiring the **rotor-spinor** ψ to satisfy $\psi\tilde{\psi} = 1$. Then, the equation (3) reduces to

$$\psi_{\pm} \mathbf{e}_3 \tilde{\psi}_{\pm} = \pm \hat{\mathbf{k}}, \quad (4)$$

where $\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$. Thus, according to the geometric algebra the eigenspinors $\psi_{\pm}$ of a monolayer graphene are rotors that bring the unit vector $\mathbf{e}_3$ that is parallel to quantization axis to new direction $\pm \hat{\mathbf{k}}$, which is either parallel or antiparallel to the electron wave vector $\mathbf{k}$.



No such clear-cut geometrical interpretation exists for eigenspinors calculated in the Hilbert space.

$Cl_{3,0}$ algebra approach

Eigenmultivectors of monolayer graphene

The solution of (4) can be written automatically as a rotor in the unit bivector $\hat{\mathbf{k}} \wedge \mathbf{e}_3$ plane,

$$\psi_{\pm} = e^{\pm \hat{\mathbf{k}} \wedge \mathbf{e}_3 \pi/4} = \frac{1}{\sqrt{2}} (1 \pm \hat{\mathbf{k}} \wedge \mathbf{e}_3). \quad (5)$$

Since $\hat{\mathbf{k}}$ and $\mathbf{e}_3$ are mutually perpendicular the outer product may be replaced by geometric one, $\hat{\mathbf{k}} \wedge \mathbf{e}_3 = \hat{\mathbf{k}} \mathbf{e}_3$. In the expanded coordinate form the eigenspinors are

$$\psi_{\pm} = \frac{1}{\sqrt{2}} (1 \pm \sin \varphi I \mathbf{e}_1 \mp \cos \varphi I \mathbf{e}_2). \quad (6)$$

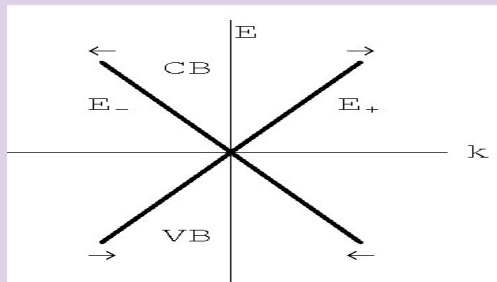
The eigenspinors (5) are written in a coordinate-free way. The appearance of $\mathbf{e}_3$ in (5) is indispensable since the quantum mechanics requires to indicate the direction of the pseudospin (or spin) quantization axis which in our case coincides with $\mathbf{e}_3$.

$Cl_{3,0}$ algebra approach

Eigenmultivectors of monolayer graphene

The spin is an intrinsic property of electron, while the pseudospin is the property of interacting p -orbitals of carbon atoms that make up the lattice. The average pseudospin vector may be found in exactly the same way as the electron spin [Doran and Lasenby, 2003],

$$\langle \mathbf{P} \rangle = \psi_{\pm} \mathbf{e}_3 \tilde{\psi}_{\pm} = \pm \mathbf{k} / |\mathbf{k}| = \pm (\cos \varphi \mathbf{e}_1 + \sin \varphi \mathbf{e}_2). \quad (7)$$



The eigenstates the pseudospin is either parallel or antiparallel to electron propagation direction.

The algorithm

Important steps

- Using replacement rules rewrite the hamiltonian in terms of geometric algebra
- Put obtained equation into rotor equation form
- Solve algebraic equation for eigenenergy
- Substitute eigenenergy into rotor equation and solve for eigenmultivector.

Test how the algorithm works for other $Cl_{3,0}$ algebra problem, i. e. for electron in quantum well.

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$Cl_{3,0}$ algebra approach

Electron in quantum well

Quantum well Hamiltonian with spin-orbit interaction α_R

$$\hat{H}_{QW} = \frac{\hbar^2}{2m^*}(k_x^2 + k_y^2)\hat{1} + \alpha_R(k_y\hat{\sigma}_x - k_x\hat{\sigma}_y). \quad (8)$$

Hamiltonian mapped into $Cl_{3,0}$ algebra

$$H_{QW}(\psi) = \frac{1}{2}(k_x^2 + k_y^2)\psi + \alpha_R(k_y\mathbf{e}_1 - k_x\mathbf{e}_2)\psi\mathbf{e}_3 = \frac{k^2}{2} + \alpha_R\mathbf{e}_{12}\mathbf{k}\psi\mathbf{e}_3. \quad (9)$$

Multivector equation, where E is the eigenenergy:

$$\frac{k^2}{2}\psi + \alpha_R\mathbf{e}_{12}\mathbf{k}\psi\mathbf{e}_3 = E\psi. \quad (10)$$

The equation rearranged into rotor form

$$\psi\mathbf{e}_3\tilde{\psi} = \frac{k^2/2 - E}{\alpha_R k^2}\mathbf{k}\mathbf{e}_{12}, \quad (11)$$

where $\mathbf{k}\mathbf{e}_{12} = \mathbf{k} \cdot \mathbf{e}_{12}$.

$Cl_{3,0}$ algebra approach

Electron in quantum well

Spectrum

$$E_{\pm} = \frac{k^2}{2} \pm k\alpha_R. \quad (12)$$

Insertion of eigenenergies $E_{\pm}$ back into (11) gives the rotor equation

$$\psi_{\pm} \mathbf{e}_3 \tilde{\psi}_{\pm} = \pm(k_y \mathbf{e}_1 - k_x \mathbf{e}_2)/k \equiv \hat{\mathbf{a}}_{\pm}. \quad (13)$$

Since $\mathbf{e}_3$ is perpendicular to quantum well plane and $\hat{\mathbf{a}}_{\pm}$ lies in the quantum well plane the rotation angle $\theta = \pm\pi/2$. Therefore, the solution of equation (13) is the rotor

$$\psi_{\pm} = \exp\left(\pm \frac{\mathbf{e}_3 \wedge \hat{\mathbf{a}}_{\pm}}{|\mathbf{e}_3 \wedge \hat{\mathbf{a}}_{\pm}|} \frac{\pi}{4}\right). \quad (14)$$

In the expanded coordinate form the eigenspinors-rotors are

$$\psi_{\pm} = (1 \mp \cos \varphi \mathbf{e}_{23} \mp \sin \varphi \mathbf{e}_{31})/\sqrt{2}, \quad (15)$$

where $\cos \varphi = k_x/k$ and $\sin \varphi = k_y/k$.

- 1 Introduction to geometric algebra
- 2 Standard quantum mechanics
- 3 Geometric algebra approach: $Cl_{3,0}$ case
 - How to dispense with matrices
 - Two level model
 - Example of monolayer graphene
 - The algorithm
 - One more example: electron in quantum well
- 4 Geometric algebra approach: $Cl_{3,1}$ case**
 - A pair of coupled two-level atoms
 - Bilayer graphene
- 5 General matrix algebras
- 6 Conclusions

$Cl_{3,1}$ algebra case

A pair of coupled two-level atoms

Two-level **single atom** standard Hamiltonian and its dipole momentum operator

$$\hat{H}_1 = \frac{\hbar\omega}{2}\hat{\sigma}_z, \quad \hat{\mathbf{d}}_1 = \boldsymbol{\mu}\hat{\sigma}_x, \quad (16)$$

where $\hbar\omega$ is the energy difference between the energy levels and $\boldsymbol{\mu}$ is the coupling constant. After mapping to the $Cl_{3,0}$ the Hamiltonian and dipole moment become

$$H_1(\psi) = \frac{\hbar\omega}{2}\mathbf{e}_3\psi\mathbf{e}_3, \quad \mathbf{d}_1(\psi) = \boldsymbol{\mu}\mathbf{e}_1\psi\mathbf{e}_3. \quad (17)$$

Now take **a pair of two-level atoms** coupled to each other through dipole-dipole coupling which is proportional to $\Gamma = \boldsymbol{\mu}^{(1)} \cdot \boldsymbol{\mu}^{(2)}$. This interaction may produce entanglement!

$Cl_{3,1}$ algebra case

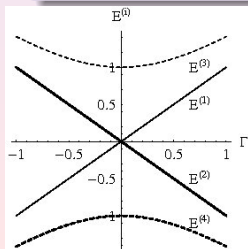
A pair of coupled two level atoms

The coupling constant Γ then depends on angle θ between the vectors $\boldsymbol{\mu}^{(1)}$ and $\boldsymbol{\mu}^{(2)}$.

The composite Hamiltonian of two two-level atoms is

$$\hat{H}_2 = \frac{\hbar\omega}{2} \left(\hat{\sigma}_z^{(1)} \otimes \hat{1} + \hat{1} \otimes \hat{\sigma}_z^{(2)} \right) + \Gamma \hat{\sigma}_x^{(1)} \otimes \hat{\sigma}_x^{(2)}. \quad (18)$$

The eigenvalues of this Hamiltonian are four energy levels, $\pm\Gamma$ and $\pm\sqrt{\Gamma^2 + \omega^2}$



$Cl_{3,0}$ algebra is too small for description of the four level system. $Cl_{3,1}$ will be used.

$Cl_{3,1}$ algebra

$Cl_{3,1}$ matrix representation

Basis vectors e_i satisfy

$$e_1^2 = e_2^2 = e_3^2 = 1 \text{ and } e_4^2 = -1.$$

Complex 4×4 matrix representation of e_i used in decomposition the Hamiltonian, pseudospin and velocity matrices

$$\hat{e}_1 = \begin{bmatrix} 0 & \hat{1} \\ \hat{1} & 0 \end{bmatrix}, \quad \hat{e}_2 = i \begin{bmatrix} 0 & -\hat{1} \\ \hat{1} & 0 \end{bmatrix},$$

$$\hat{e}_3 = \begin{bmatrix} \hat{\sigma}_y & 0 \\ 0 & -\hat{\sigma}_y \end{bmatrix}, \quad \hat{e}_4 = i \begin{bmatrix} \hat{\sigma}_z & 0 \\ 0 & -\hat{\sigma}_z \end{bmatrix},$$

where $i = \sqrt{-1}$. $\hat{1}$ is 2×2 unit matrix and $\hat{\sigma}_x$, $\hat{\sigma}_y$, $\hat{\sigma}_z$ are Pauli matrices.

Replacement rules

The following mapping is assumed between the complex spinor $|\psi\rangle$ in a form of column and GA spinor ψ ,

$$|\psi\rangle = \begin{bmatrix} a_0 + ia_3 \\ -b_3 + ib_0 \\ -b_2 - ib_1 \\ -a_1 + ia_2 \end{bmatrix}$$

$\Updownarrow$

$$\begin{aligned} \psi &= a_0 + a_1 e_{23} - a_2 e_{31} \\ &\quad + a_3 e_{12} - b_0 I - b_1 e_{14} \\ &\quad + b_2 e_{24} + b_3 e_{34}, \end{aligned} \quad (19)$$

where a_i 's and b_i 's are scalars.

$Cl_{3,1}$ algebra case

Replacement rules (continued)

The knowledge of matrix representation of basis elements and spinor mapping rule (19) allow to construct the main replacement rules between the action of matrices on column vectors and $Cl_{3,1}$ multivector functions [Dargys, 2014a, Dargys, 2014b],

$$\begin{aligned}\hat{e}_i|\psi\rangle &\Longleftrightarrow \mathbf{e}_i\psi\mathbf{Ie}_3, & \mathbf{i}|\psi\rangle &\Longleftrightarrow I\psi\mathbf{e}_{34}, \\ \hat{e}_{ij}|\psi\rangle &\Longleftrightarrow \mathbf{e}_{ij}\psi, & \langle\psi|\psi\rangle &\Longleftrightarrow \langle\psi^\dagger\psi\rangle, \\ \hat{I}\hat{e}_i|\psi\rangle &\Longleftrightarrow \mathbf{Ie}_i\psi\mathbf{Ie}_3, & \langle\varphi|\psi\rangle &\Longleftrightarrow \langle\varphi^\dagger\psi\rangle - \langle\varphi^\dagger\psi\mathbf{e}_{12}\rangle\mathbf{e}_{12}\end{aligned}$$

where $i, j = 1, 2, 3, 4$. The dagger operation is a combination of the reversion and spatial inversion

$$\psi^\dagger = -\mathbf{e}_4\tilde{\psi}\mathbf{e}_4.$$

If applied to a general bispinor it allows to find the square of the module

$$\langle\psi^\dagger\psi\rangle = a_0^2 + a_1^2 + a_2^2 + a_3^2 + b_0^2 + b_1^2 + b_2^2 + b_3^2.$$

$Cl_{3,1}$ algebra case

A pair of coupled two level atoms (continued)

After mapping to $Cl_{3,1}$ the **Hamiltonian of a pair of two-level atoms** becomes

$$H_2(\psi) = \frac{\omega}{2} \mathbf{e}_{34} \psi \mathbf{e}_{34} - \Gamma \mathbf{e}_2 \psi \mathbf{e}_3 + \frac{\omega}{2} \mathbf{e}_3 \psi \mathbf{e}_3. \quad (20)$$

The first term requires double application of the replacement rules: $i\hat{e}_{12}|\psi\rangle \rightarrow I(\mathbf{e}_{12}\psi)\mathbf{e}_{34} = -\mathbf{e}_{34}\psi\mathbf{e}_{34}$. To find the eigens of the equation $H_2(\psi) = E\psi$ we need spatial inversion operation $\bar{\psi} = -\mathbf{e}_4\psi\mathbf{e}_4$. With help of it the spinor can be written as a sum of even $\bar{\psi}_+ = \psi_+$ and odd $\bar{\psi}_- = -\psi_-$ parts. If the spatial inversion is applied, the Hamiltonian (20) and its inverse become

$$\begin{aligned} H_2(\psi) &= \omega \mathbf{e}_3 \psi_- \mathbf{e}_3 - \Gamma \mathbf{e}_2 (\psi_+ + \psi_-) \mathbf{e}_3, \\ \bar{H}_2(\psi) &= -\omega \mathbf{e}_3 \psi_- \mathbf{e}_3 - \Gamma \mathbf{e}_2 (\psi_+ - \psi_-) \mathbf{e}_3. \end{aligned} \quad (21)$$

$Cl_{3,1}$ algebra case

A pair of coupled two level atoms

Since the eigenequation $H_2(\psi) = E\psi$ and its inversion $\bar{H}_2(\psi) = E\bar{\psi}$ give $(H_2(\psi) + \bar{H}_2(\psi))/2 = E\psi_+$ and $(H_2(\psi) - \bar{H}_2(\psi))/2 = E\psi_-$, we find that equations for even and odd states **decouple**

$$-\Gamma \mathbf{e}_2 \psi_+ \mathbf{e}_3 = E \psi_+, \quad (22)$$

$$\omega \mathbf{e}_3 \psi_- \mathbf{e}_3 - \Gamma \mathbf{e}_2 \psi_- \mathbf{e}_3 = E \psi_-. \quad (23)$$

Assuming that $\psi_+ \tilde{\psi}_+ = 1$, the first equation (22) can be put into the **rotor form**

$$\psi_+ \mathbf{e}_3 \tilde{\psi}_+ = -\frac{E}{\Gamma} \mathbf{e}_2, \quad (24)$$

from which we find the dispersion

$$E^{(1,2)} = \pm \Gamma \quad (25)$$

and rotor equation

$$\psi_+ \mathbf{e}_3 \tilde{\psi}_+ = \mp \mathbf{e}_2. \quad (26)$$

$Cl_{3,1}$ algebra case

A pair of coupled two level atoms

The solutions of this equation are the rotors

$$\psi_+^{(1,2)} = e^{\pm \mathbf{e}_{23}\pi/4} = \frac{1}{\sqrt{2}}(1 \pm \mathbf{e}_{23}). \quad (27)$$

The first rotor $\psi_+^{(1)}$ gives the eigenenergy $E_+^{(1)} = \langle \tilde{\psi}_+^{(1)} \mathbf{H}(\psi_+^{(1)}) \rangle = -\Gamma$ and rotates $\mathbf{e}_3$ to $\mathbf{e}_2$, while $\psi_+^{(2)}$ gives $E_+^{(2)} = \langle \tilde{\psi}_+^{(2)} \mathbf{H}(\psi_+^{(2)}) \rangle = \Gamma$ and rotates $\mathbf{e}_3$ to $-\mathbf{e}_2$.

$Cl_{3,1}$ algebra case

A pair of coupled two level atoms

The second eigenequation (23) can be rearranged to the rotor form

$$\psi_{-\mathbf{e}_3} \tilde{\psi}_{-} = \frac{E(\omega \mathbf{e}_3 - \Gamma \mathbf{e}_2)}{\omega^2 + \Gamma^2}. \quad (28)$$

If equated to 1, the square of the right hand side gives the dispersion

$$E^{(2,3)} = \pm \sqrt{\omega^2 + \Gamma^2}. \quad (29)$$

After insertion of $E^{(2,3)}$ into eigenequation and rearrangement one gets the following **rotor equation**

$$\psi_{-\mathbf{e}_3} \tilde{\psi}_{-} = \pm \frac{\omega \mathbf{e}_3 - \Gamma \mathbf{e}_2}{\sqrt{\omega^2 + \Gamma^2}} \equiv \hat{\mathbf{a}}. \quad (30)$$

Solution of this equation is

$$\psi = e^{\frac{\mathbf{e}_3 \wedge \hat{\mathbf{a}}}{|\mathbf{e}_3 \wedge \hat{\mathbf{a}}|} \frac{\theta}{2}} = \cos \frac{\theta}{2} \pm \frac{\mathbf{e}_3 \wedge \hat{\mathbf{a}}}{|\mathbf{e}_3 \wedge \hat{\mathbf{a}}|} \sin \frac{\theta}{2}, \quad (31)$$

where $\cos \theta = \mathbf{e}_3 \cdot \hat{\mathbf{a}}$. In our case $\cos \theta = \omega / \sqrt{\omega^2 + \Gamma^2}$ and $\frac{\mathbf{e}_3 \wedge \hat{\mathbf{a}}}{|\mathbf{e}_3 \wedge \hat{\mathbf{a}}|} = \mathbf{e}_{23}$, therefore, the rotation acts in the $\mathbf{e}_{23}$ plane.

$Cl_{3,1}$ algebra case

Bilayer graphene

In the previous example the rotors ψ_+ and ψ_- were decoupled what greatly simplifies the problem. Here we shall consider more complex case of bilayer graphene (BG) where **rotor equations are coupled**. Using matrix representation of $Cl_{3,1}$ algebra the BG Hamiltonian can be decomposed into sum of matrices that represent $Cl_{3,1}$

$$\hat{H}_{BG} = \eta(k_x \hat{e}_1 + k_y \hat{e}_2) - \frac{\gamma_1}{2} (\hat{e}_3 \hat{e}_1 + \hat{I} \hat{e}_2) + \eta U \hat{I} \hat{e}_3. \quad (32)$$

γ_1 is the interlayer coupling energy (constant). The energy $2U$ is the potential difference between the layers. $\eta = +1$ for K -valley and $\eta = -1$ for K' -valley. The electron spin is not included in this Hamiltonian. All energies are normalized to intralayer hopping energy γ_0 . The wave vector and its components will be normalized by $k_c = 2\pi/a$, where a is the graphene lattice constant. Also we have assumed that $\hbar = 1$.

$Cl_{3,1}$ algebra case

Bilayer graphene

Using the replacement rules the matrix Hamiltonian (32) can be mapped to GA Hamiltonian

$$H_{BG}(\psi) = \eta \mathbf{k} \psi \mathbf{I} \mathbf{e}_3 - \frac{\gamma_1}{2} \mathbf{e}_2 (\psi - \bar{\psi}) \mathbf{e}_3 + \eta U \mathbf{e}_3 \psi \mathbf{e}_3, \quad (33)$$

where $\mathbf{k} = k_x \mathbf{e}_1 + k_y \mathbf{e}_2$ is the in-plane wave vector of electron.

To transform the eigenequation $H_{BG}(\psi) = E\psi$ to rotors the spinor is split into even and odd parts, $\psi = \psi_+ + \psi_-$, and then the eigenequation is decomposed into two **coupled** multivector equations for ψ_+ and ψ_- ,

$$\begin{aligned} E\psi_+ &= \eta \mathbf{k} \psi_- \mathbf{I} \mathbf{e}_3 + \eta U \mathbf{e}_3 \psi_+ \mathbf{e}_3, \\ E\psi_- &= \eta \mathbf{k} \psi_+ \mathbf{I} \mathbf{e}_3 + \eta U \mathbf{e}_3 \psi_- \mathbf{e}_3 - \gamma_1 \mathbf{e}_2 \psi_- \mathbf{e}_3. \end{aligned} \quad (34)$$

The method of solution of the system (34) uses the fact that ψ_+ and $I\psi_-$, where I is the pseudoscalar of $Cl_{3,1}$ algebra, are the rotors in 3D Euclidean space. For this purpose ψ_- is expressed from the first equation and inserted into the second equation of (34).

$Cl_{3,1}$ algebra case

Bilayer graphene

After algebraic manipulations one can construct the following **rotor equation** for ψ_+ ,

$$\psi_+ \mathbf{e}_3 \tilde{\psi}_+ = NM^{-1} \equiv \hat{\mathbf{a}}, \quad (35)$$

where

$$\begin{aligned} M &= E[2\eta U I \mathbf{e}_3 - k^{-2} \gamma_1 \mathbf{e}_1 (k_x + k_y \mathbf{e}_{12})^2 \mathbf{e}_{34}], \\ N &= (E^2 + U^2 - k^2)I + k^{-2} \eta U \gamma_1 \mathbf{e}_1 (k_x + k_y \mathbf{e}_{12})^2 \mathbf{e}_4. \end{aligned} \quad (36)$$

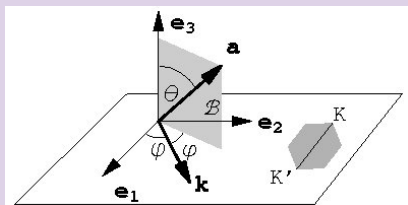
The unit vector $\hat{\mathbf{a}}$ can be simplified to the following form

$$\begin{aligned} \hat{\mathbf{a}} &= \frac{\gamma_1 \xi}{E k^2} \mathbf{k} \mathbf{e}_2 \mathbf{k} + \frac{\eta U (2\xi + 1)}{E} \mathbf{e}_3, \\ \xi &= \frac{E^2 - k^2 - U^2}{4U^2 + \gamma_1^2}. \end{aligned} \quad (37)$$

The geometric product $\mathbf{k} \mathbf{e}_2 \mathbf{k}$ of three vectors in polar coordinates simplifies to $\mathbf{k} \mathbf{e}_2 \mathbf{k} / k^2 = \sin(2\varphi) \mathbf{e}_1 - \cos(2\varphi) \mathbf{e}_2$, where φ is the angle between $\mathbf{e}_1$ and $\mathbf{k}$.

$Cl_{3,1}$ algebra case

Geometric interpretation of bilayer graphene equation (35)



The initial vector $\mathbf{e}_3$, which is perpendicular to graphene plane and coincides with the quantization axes, is brought to the final vector $\hat{\mathbf{a}}$ which is determined by BG Hamiltonian. In general case, the angle θ depends on the competition between the interlayer interaction strength γ_1 and the external voltage $2U$ applied between the layers.

Two extreme cases

- Interlayer interaction is switched off, i. e. $\gamma_1 = 0$, then $\hat{\mathbf{a}} \parallel \mathbf{e}_3$ and the rotations can be done around $\mathbf{e}_3$ axis only.
- When $U = 0$, the final vector $\hat{\mathbf{a}}$ lies in the graphene plane, so that the rotation angle in this case is $\theta = \pi/2$.

$Cl_{3,1}$ algebra case

Bilayer graphene spectrum

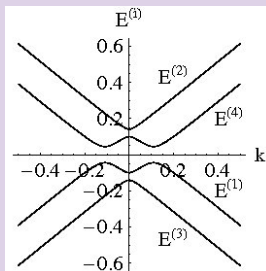
The requirement that the final vector $\mathbf{a}$ after rotation (35) will not change its length is expressed by condition $\hat{\mathbf{a}}^2 = 1$,

$$\hat{\mathbf{a}}^2 = \frac{(E^2 - k^2 + U^2)^2 + U^2 \gamma_1^2}{E^2(4U^2 + \gamma_1^2)} = 1. \quad (38)$$

This fourth order polynomial equation gives BG spectrum

$$E^{(i)} = \pm \sqrt{k^2 + U^2 + \frac{\gamma_1^2}{2}} \pm \frac{1}{2} \sqrt{\gamma_1^4 + 4k^2(4U^2 + \gamma_1^2)}, \quad (39)$$

where $k = (k_x^2 + k_y^2)^{1/2}$ and i is the band number as indicated in the figure.



The spectrum (39) is symmetric with respect to k axis and is same for both K and K' valleys. Two of the bands have “Mexican-hat” structure that brings altogether new physical properties. When $U = 0$ the “Mexican-hat” structure and energy gap vanishes, and the spectrum reduces to $E^{(i)} = \pm(\gamma_1/2) \pm \sqrt{k^2 + (\gamma_1/2)^2}$.

$Cl_{3,1}$ algebra case

Solutions of bilayer graphene equation

Knowing $\hat{\mathbf{a}}^{(i)} = \mathbf{a}^{(i)} / |\mathbf{a}^{(i)}|$ and $E^{(i)}$, one can construct the rotor for the i -th energy band. The cosine of the angle $\theta^{(i)}$ between $\mathbf{e}_3$ and $\hat{\mathbf{a}}^{(i)}$ can be found from the inner GA product $\cos \theta^{(i)} = \mathbf{e}_3 \cdot \hat{\mathbf{a}}^{(i)}$,

$$\cos \theta^{(i)} \equiv c^{(i)} = \eta \frac{U(2E^{(i)2} - 2k^2 + 2U^2 + \gamma_1^2)}{\sqrt{4U^2 + \gamma_1^2} \sqrt{(E^{(i)2} - k^2 + U^2)^2 + U^2 \gamma_1^2}}. \quad (40)$$

Finally, elementary but lengthy algebraic calculations allow to give to the rotor-spinor the following form

$$\psi_+^{(i)} = \frac{1}{\sqrt{2}} \left(\sqrt{1 + c^{(i)}} + (1 - c^{(i)}) \frac{\mathbf{e}_3 \mathbf{k} \mathbf{e}_2 \mathbf{k}}{k^2} \right). \quad (41)$$

After expansion one can show that $\psi_+^{(i)}$ consists of scalar and two bivectors, $\mathbf{e}_{23}$ and $\mathbf{e}_{31}$.

$Cl_{3,1}$ algebra case

Solutions of bilayer graphene equation

The knowledge of $\psi_+^{(i)}$ allows to find the second spinor-rotor $(I\psi_-^{(i)})$ which is related with the odd part of ψ . It can be expressed through ψ_+ if the first equation of the system (34) is used,

$$(I\psi_-^{(i)}) = \frac{\mathbf{k}}{k^2} \left(\eta E^{(i)} \psi_+^{(i)} \mathbf{e}_3 - U \mathbf{e}_3 \psi_+^{(i)} \right), \quad (42)$$

which can be rearranged to the following form

$$\psi_-^{(i)} = \frac{\sqrt{1+c^{(i)}}}{\sqrt{2}k} \left(U - E^{(i)}\eta \right) e^{-\mathbf{e}_{12}\varphi} \mathbf{e}_{24} + \frac{\sqrt{1-c^{(i)}}}{\sqrt{2}k} \left(U + E^{(i)}\eta \right) e^{\mathbf{e}_{12}\varphi} \mathbf{e}_{34}. \quad (43)$$

The total eigenmultivector $\psi^{(i)}$ is a sum of (41) and (43),

$$\psi^{(i)} = \psi_+^{(i)} + \psi_-^{(i)}. \quad (44)$$

In calculating the physical averages the eigenmultivector $\psi^{(i)}$ must be normalized.

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General matrix algebras

$Cl_{p,q}$ matrix representations

$p \backslash q$	0	1	2	3	4	5	6	7	...
0	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	${}^2\mathbb{H}$	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$	...
1	${}^2\mathbb{R}$	$\mathbb{R}(2)$	$\mathbb{C}(2)$	$\mathbb{H}(2)$	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$	...
2	$\mathbb{R}(2)$	${}^2\mathbb{R}(2)$	$\mathbb{R}(4)$	$\mathbb{C}(4)$	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$	...
3	$\mathbb{C}(2)$	$\mathbb{R}(4)$	${}^2\mathbb{R}(4)$	$\mathbb{R}(8)$	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$	...
4	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$	$\mathbb{R}(16)$	$\mathbb{C}(16)$	$\mathbb{H}(16)$	${}^2\mathbb{H}(16)$	...
5	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$	${}^2\mathbb{R}(16)$	$\mathbb{R}(32)$	$\mathbb{C}(32)$	$\mathbb{H}(32)$	...
6	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$	$\mathbb{R}(32)$	${}^2\mathbb{R}(32)$	$\mathbb{R}(64)$	$\mathbb{C}(64)$	...
7	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$	$\mathbb{C}(32)$	$\mathbb{R}(64)$	${}^2\mathbb{R}(64)$	$\mathbb{R}(128)$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Eightfold periodic table of $Cl_{p,q}$ Clifford algebras. $\mathbb{R}(n)$, $\mathbb{C}(n)$ and $\mathbb{H}(n)$ represents $n \times n$ real, complex and quaternionic matrix. Symbol ${}^2\mathbb{F}$ denotes direct sum of matrices $\mathbb{F} \oplus \mathbb{F}$

General matrix algebras

$Cl_{p,q}$ algebras applications

Geometric algebra	Dimension n of vector space	Area of application
$Cl_{0,1}$	1D	Complex numbers
$Cl_{2,0}$	2D	Euclidean plane
$Cl_{0,2}$	2D	Quaternions
$Cl_{3,0}$	3D	Euclidean space
	3D	Projective geometry, animation
	3D	Quantum mechanics, electrodynamics
$Cl_{3,1}$	4D	Relativistic spacetime, Dirac equation
	4D	relativistic electrodynamics, gravitation
$Cl_{4,1}$	5D	Graphics, animation
	5D	conformal geometry

Some of application of the most popular geometric (Clifford) algebras $Cl_{p,q}$ in $n = p + q$ dimensional space.

The only note that the **mysterious imaginary unit** $i = \sqrt{-1}$ is nothing else but a bivector generator which describes rotations in the Euclidean plane.

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- Presence of a specific direction called the quantization axis in the Hamiltonian can be employed to solve quantum problems in GA.
- Interpretation of the wave function as GA rotor can be used to solve quantum mechanical eigenproblems.
- Since GA approach is coordinate-free the formulas found in this way appear to be rather compact and may be interpreted geometrically.
- The method was demonstrated by a number of examples from physics: monolayer and bilayer graphene, two-level atoms, quantum wells.

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Thank You